

Heart Disease Prediction: End-to-End MLOps Report

Course: Machine Learning Operations (MLOps), AIMLCZG523 **Assignment:** 01, End-to-End ML Model Development, CI/CD & Production Deployment **Dataset:** Heart Disease UCI (Cleveland), UCI Machine Learning Repository

1. Project Overview

This project delivers a complete, reproducible MLOps pipeline that predicts the risk of heart disease from patient clinical data and serves it as a monitored, cloud-ready REST API. It covers the full lifecycle: data acquisition, EDA, feature engineering, model development with experiment tracking, packaging, automated testing & CI/CD, containerization, Kubernetes deployment, and monitoring.

Problem type: binary classification (0 = no heart disease, 1 = disease present).

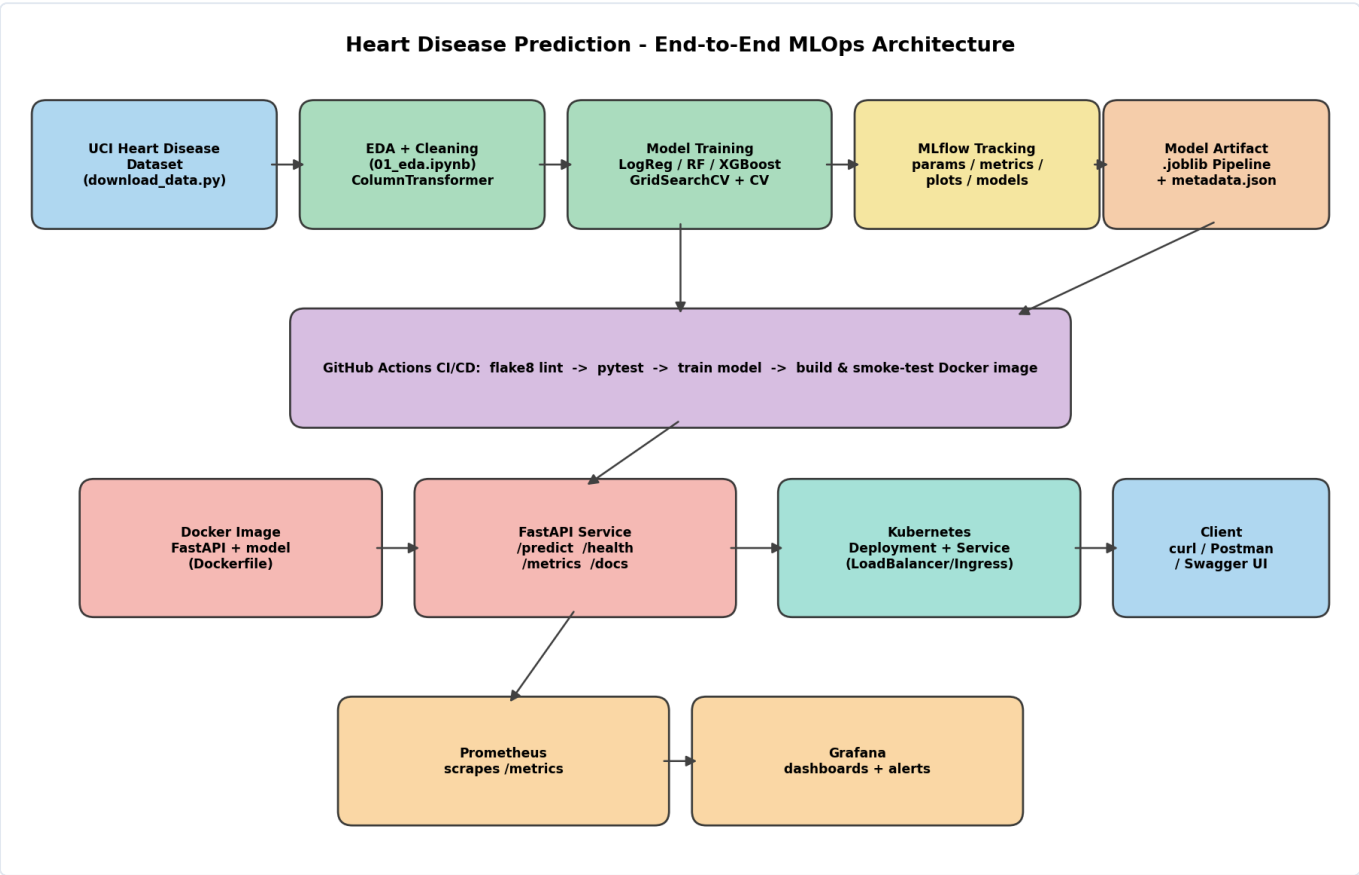


Figure 1: End-to-end MLOps architecture.

2. Data Acquisition & Exploratory Data Analysis (Task 1)

2.1 Acquisition

The dataset is obtained programmatically by `data/download_data.py`, which uses the official `ucimlrepo` package (dataset id = 45) with a direct-HTTP fallback to `processed.cleveland.data`. The raw file has **303 patients** and **14 columns** (13 clinical features + a 0-4 diagnosis column `num`).

2.2 Cleaning & preprocessing

- `?` placeholders are coerced to `NaN`. Only `ca` and `thal` contain missing values (a few rows each), which are median-imputed.
- The 0-4 diagnosis `num` is converted to a **binary target** (`num > 0` becomes 1).
- Numeric features are standard-scaled; categorical clinical codes are one-hot encoded (inside the sklearn pipeline, so it is reused at inference).

2.3 EDA findings

Visualization	File	Insight
Missing-value map	screenshots/eda_missing_values.png	Only <code>ca</code> , <code>thal</code> missing
Class balance	screenshots/eda_class_balance.png	~54% / 46%, no severe imbalance
Histograms	screenshots/eda_histograms.png	Features on very different scales, scaling needed
Correlation heatmap	screenshots/eda_correlation_heatmap.png	<code>cp</code> , <code>oldpeak</code> , <code>exang</code> , <code>ca</code> , <code>thal</code> most predictive; <code>thalach</code> negatively correlated
Numeric vs target	screenshots/eda_numeric_vs_target.png	Disease patients reach lower max heart rate, higher <code>oldpeak</code>
Categorical vs target	screenshots/eda_categorical_vs_target.png	Chest-pain type and exercise-induced angina strongly separate classes

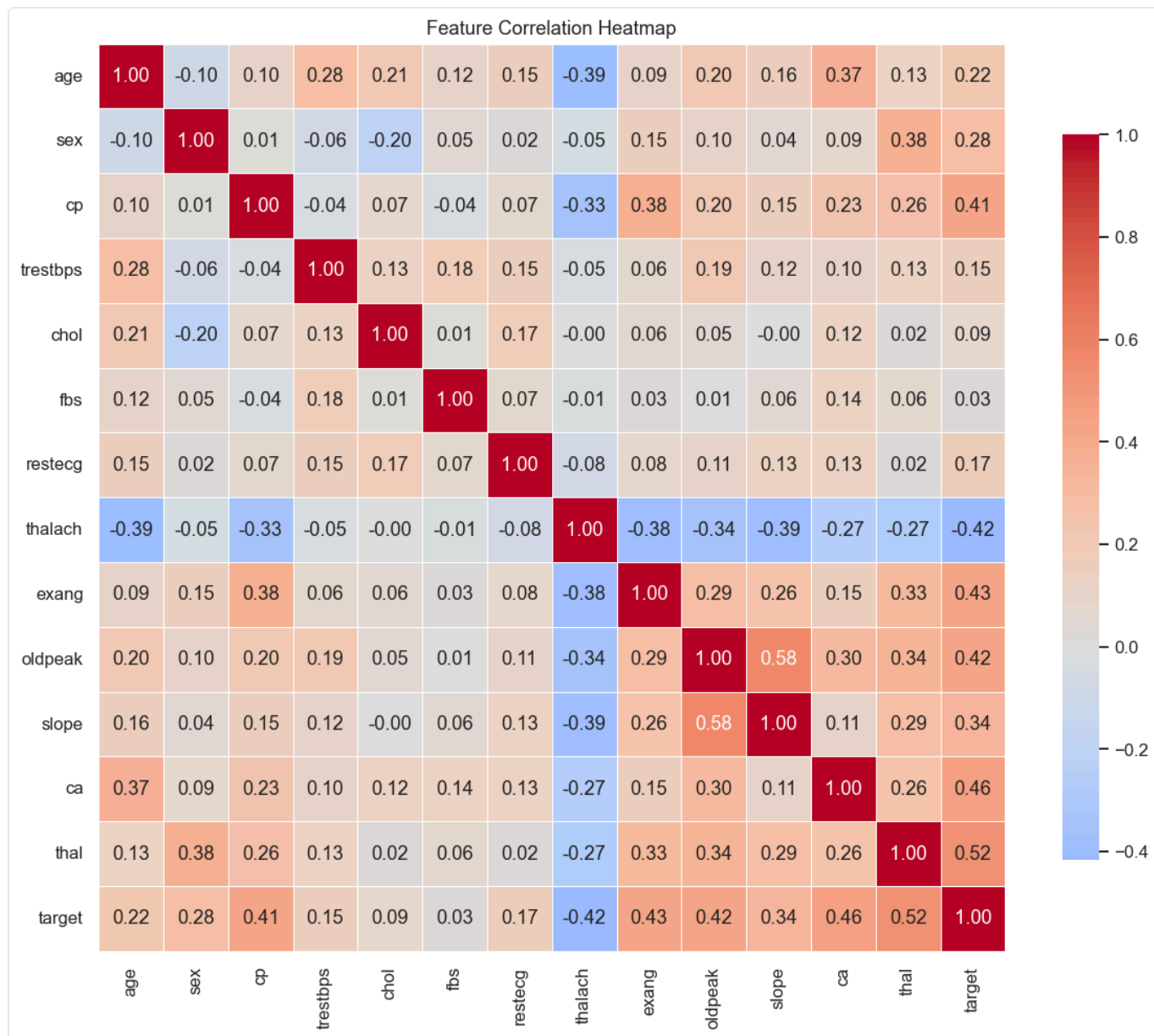


Figure 2: Feature correlation heatmap.

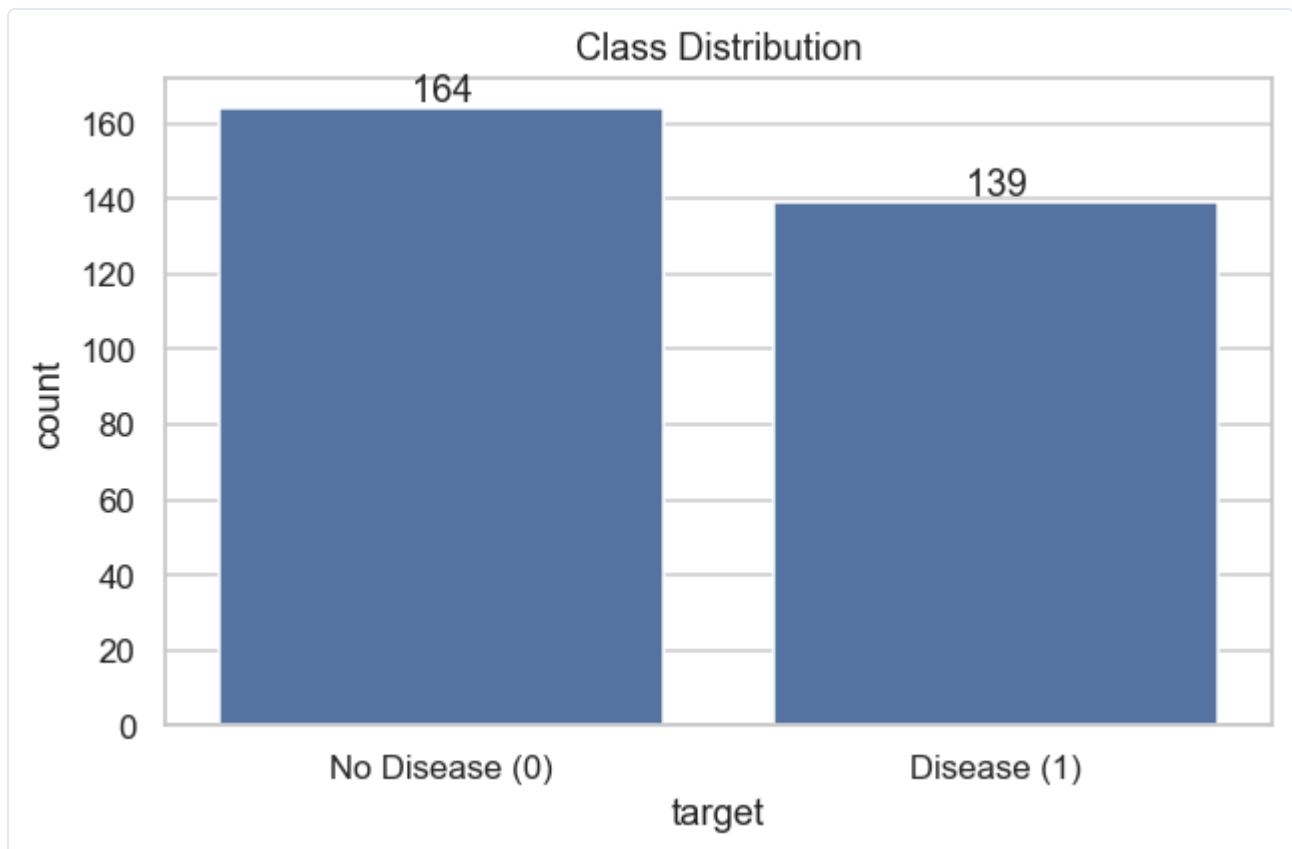


Figure 3: Target class distribution.

3. Feature Engineering & Model Development (Task 2)

3.1 Feature pipeline

A scikit-learn `ColumnTransformer` inside a `Pipeline`: * **Numeric** (age, trestbps, chol, thalach, oldpeak): median impute, then `StandardScaler`. * **Categorical** (sex, cp, fbs, restecg, exang, slope, ca, thal): most-frequent impute, then `OneHotEncoder(handle_unknown="ignore")`.

3.2 Models, tuning & evaluation

Three classifiers were trained with `GridSearchCV` (5-fold Stratified CV, scoring = ROC-AUC) and evaluated on a stratified 20% hold-out test set.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV ROC-AUC
Logistic Regression (selected)	0.885	0.839	0.929	0.881	0.966	0.902
Random Forest	0.885	0.839	0.929	0.881	0.955	0.898
XGBoost	0.902	0.867	0.929	0.897	0.931	0.870

Model selection: Logistic Regression is chosen as the production model. It has the **highest ROC-AUC (0.966)** and best cross-validated AUC (0.902), while being the simplest and most interpretable. High recall (0.93) is important clinically, since it means few missed disease cases.

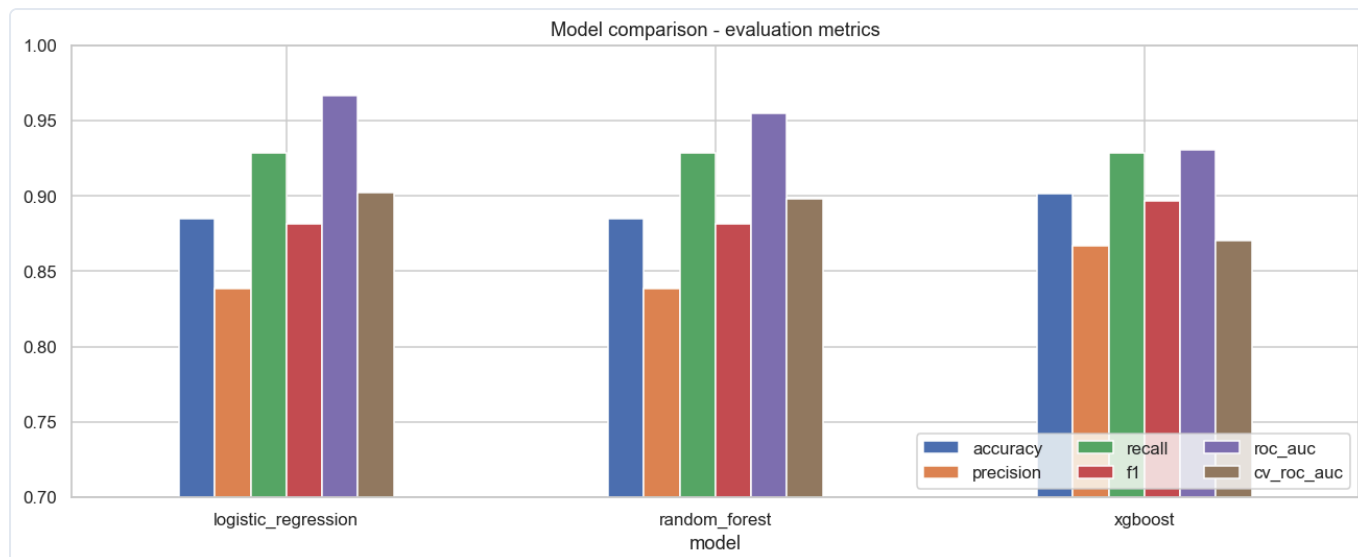


Figure 4: Model comparison across metrics.

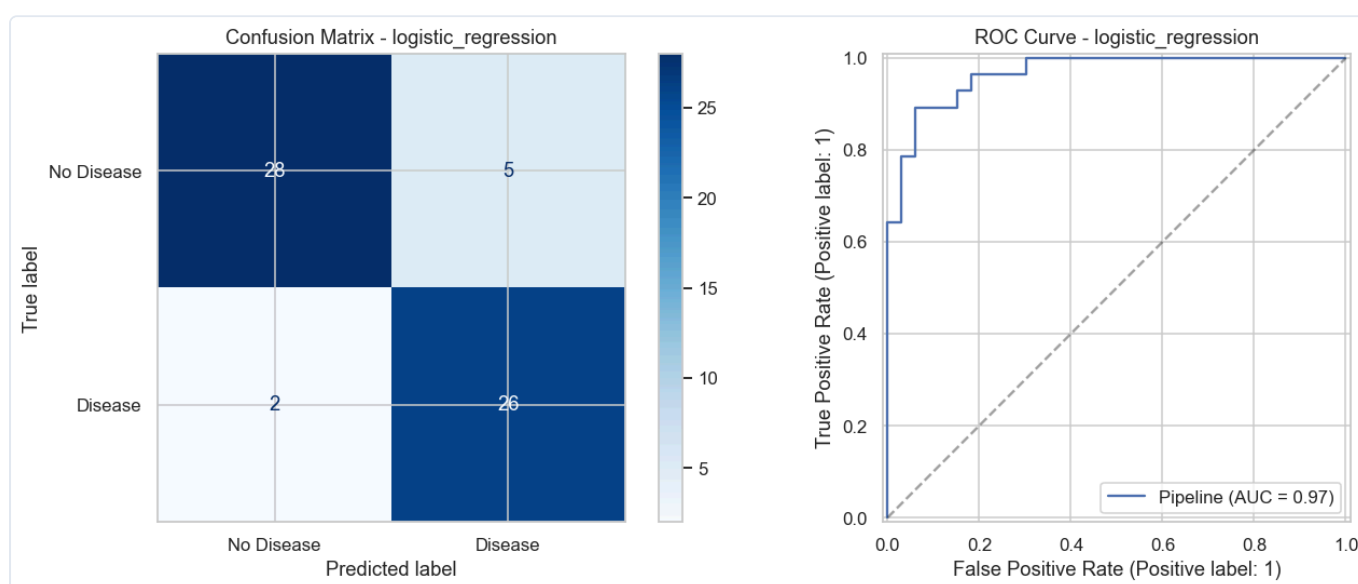


Figure 5: Confusion matrix and ROC curve for the selected model.

4. Experiment Tracking (Task 3)

MLflow tracks every run under the experiment `heart-disease-classification`. For each model we log: * Parameters: model type and best hyper-parameters from GridSearchCV. * Metrics: accuracy, precision, recall, F1, ROC-AUC, CV ROC-AUC. * Artifacts: confusion-matrix and ROC-curve PNGs. * The serialized sklearn model. Launch the UI with `mlflow ui --backend-store-uri ./mlruns` and open <http://localhost:5000>.

Experiments > heart-disease-classification >

Runs

Search: metrics.rmse < 1 and params.model = "tree" | All time | State: Active | Datasets | Sort: Created | Columns | + New run

	Run Name	Created	Duration	Source	Models	accuracy	f1	precision	recall	roc_auc
☐	xgboost	2 hours ago	0.6s	C:\Users\sajib\AppData\...	model	0.90163934426...	0.896551724...	0.866666666...	0.928571428...	0.930735930...
☐	random_forest	2 hours ago	0.6s	C:\Users\sajib\AppData\...	model	0.88524590163...	0.881355932...	0.838709677...	0.928571428...	0.954545454...
☐	logistic_regression	2 hours ago	2.8s	C:\Users\sajib\AppData\...	model	0.88524590163...	0.881355932...	0.838709677...	0.928571428...	0.966450216...

Figure 6: MLflow experiment runs and logged metrics for all three models.

The confusion-matrix and ROC-curve artifacts logged with each run are shown below.

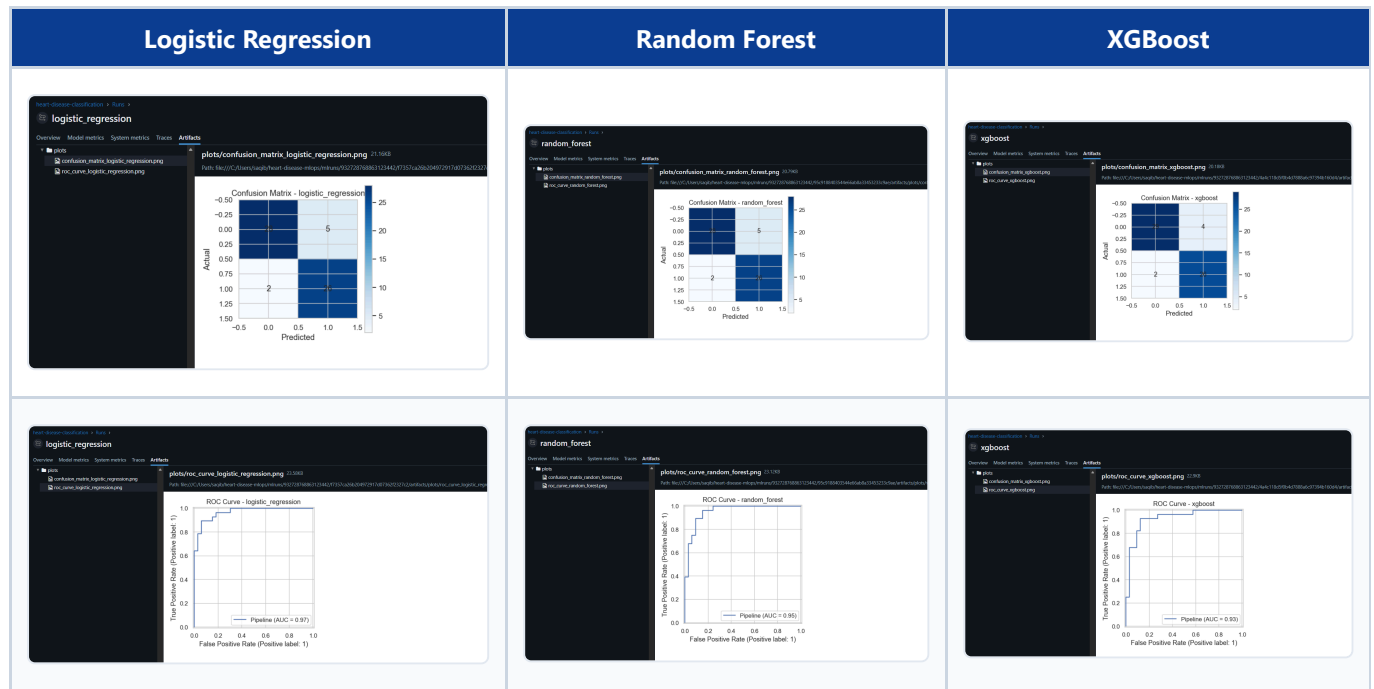


Figure 6a: Confusion matrix (top) and ROC curve (bottom) logged as MLflow artifacts for each model.

5. Model Packaging & Reproducibility (Task 4)

- The winning **Pipeline** (preprocessing + model) is serialized with `joblib` to `models/heart_disease_model.joblib`; `models/model_metadata.json` stores the chosen model, hyper-parameters, metrics and training timestamp.
- Because preprocessing lives inside the pipeline, inference requires only the single artifact, guaranteeing train/serve parity.
- Dependencies are pinned in `requirements.txt` (full) and `requirements-api.txt` (slim serving set). A fixed `RANDOM_STATE = 42` makes runs reproducible.

6. CI/CD Pipeline & Automated Testing (Task 5)

GitHub Actions (`.github/workflows/ci.yml`) runs four sequential jobs. The pipeline **fails fast** on any lint or test error:

1. **lint:** `flake8` (fails on syntax/undefined-name errors).
2. **test:** `pytest` (data, model & API tests); uploads JUnit results.
3. **train:** trains the model, uploads `trained-model` and `mlflow-runs` artifacts.
4. **docker-build:** builds the image, runs the container and smoke-tests `/health` + `/predict` .

Local test run: **13 passed** (`python -m pytest -v`).

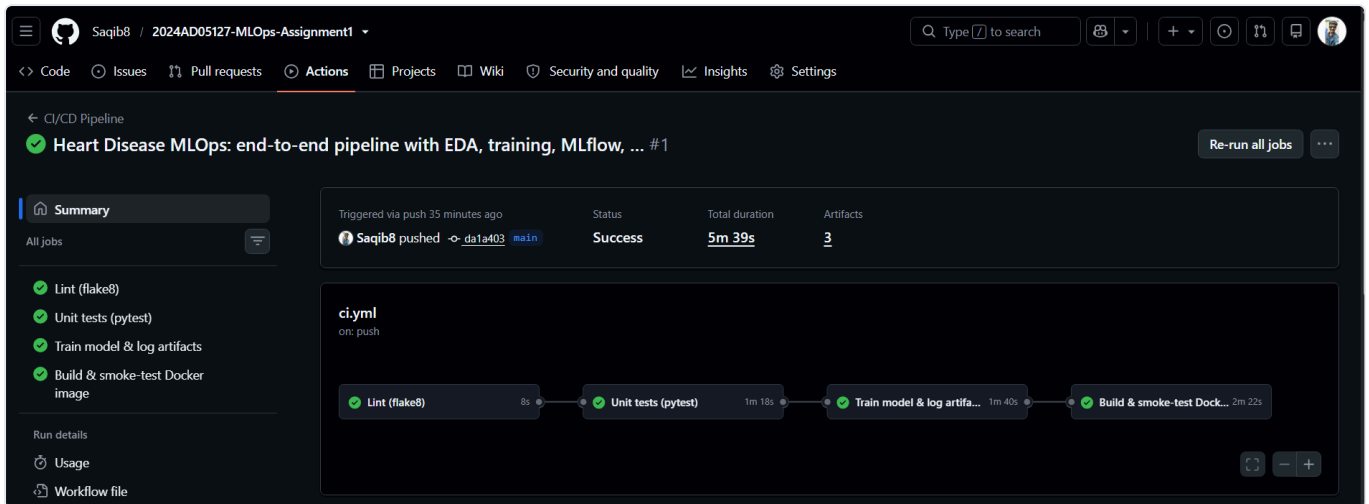


Figure 7: GitHub Actions pipeline with the four jobs passing.

7. Model Containerization (Task 6)

- Dockerfile uses python:3.12-slim, installs requirements-api.txt, copies src/, api/ and the trained models/, runs as a **non-root** user with a /health HEALTHCHECK, and serves with uvicorn on port 8000.
- The /predict endpoint accepts JSON patient features and returns the prediction, label, confidence and disease probability. Tested via the FastAPI Swagger UI (/docs): a POST /predict returns HTTP 200 with {"prediction":0, "label":"No heart disease", "confidence":0.8146, "probability_disease":0.1854} (served by uvicorn).



Figure 8: Successful POST /predict response (HTTP 200) via the API Swagger UI, returning prediction and confidence.

The same application is containerized with Docker for isolated, reproducible serving:

```
docker build -t heart-disease-api .
docker run -p 8000:8000 heart-disease-api
```

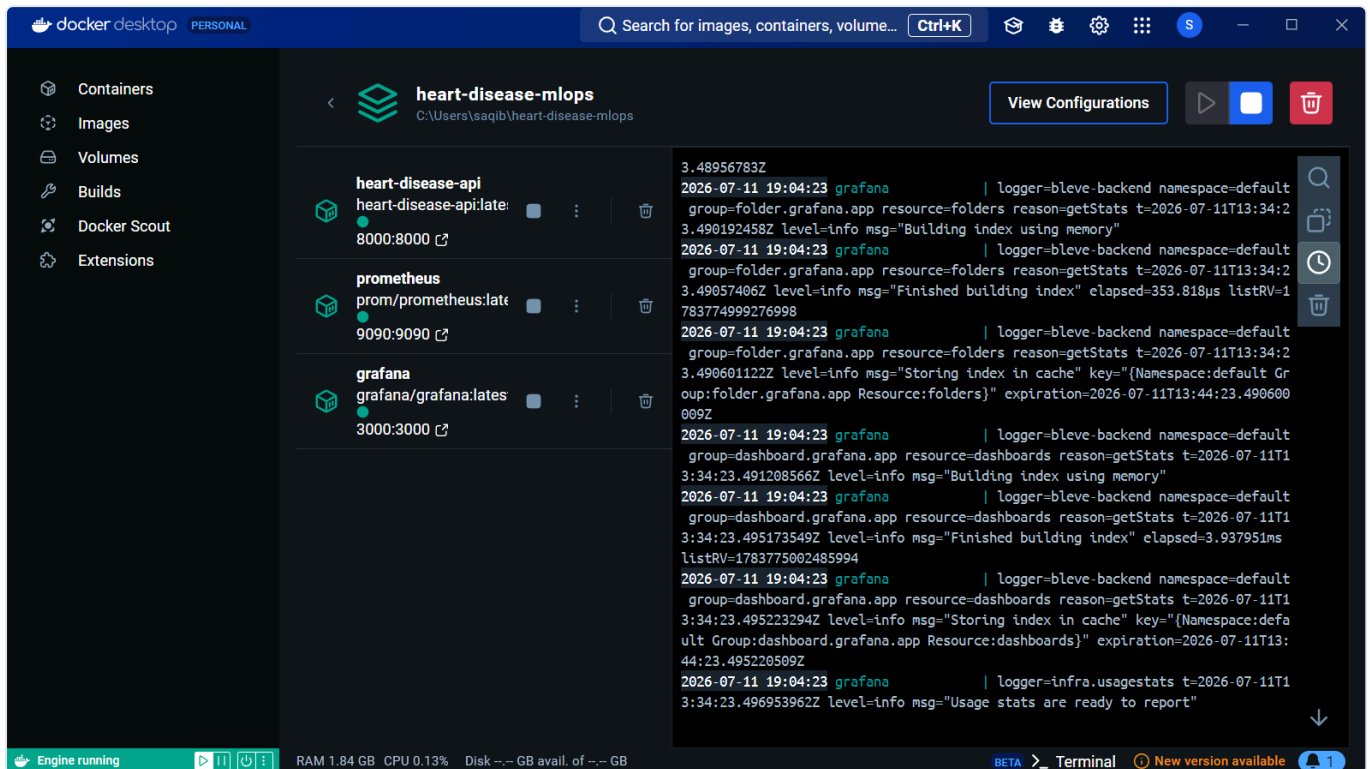


Figure 9: Docker image build and container run serving the same API.

8. Production Deployment (Task 7)

Kubernetes manifests in `k8s/` : * `deployment.yaml` : 2 replicas, resource requests/limits, liveness & readiness probes on `/health` , Prometheus scrape annotations. * `service.yaml` : LoadBalancer exposing port 80 to container 8000. * `ingress.yaml` : optional NGINX ingress (`heart-disease.local`). * `hpa.yaml` : Horizontal Pod Autoscaler (2 to 5 replicas at 70% CPU).

Deployed on Minikube / Docker Desktop Kubernetes (see README section 7).

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\saqib> cd C:\Users\saqib\heart-disease-mlops
PS C:\Users\saqib\heart-disease-mlops> kubectl get deployment,pods,svc -l app=heart-disease-api
NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/heart-disease-api    2/2      2              2             5m57s

NAME                                READY    STATUS      RESTARTS    AGE
pod/heart-disease-api-565bc5b94f-ph87q 1/1      Running     0            5m56s
pod/heart-disease-api-565bc5b94f-vnmb7 1/1      Running     0            5m56s

NAME                                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)        AGE
service/heart-disease-api            LoadBalancer  10.102.139.7   localhost      80:30166/TCP    5m56s
PS C:\Users\saqib\heart-disease-mlops> |
```

Figure 10: `kubectl get pods,svc` and a prediction against the exposed service.

9. Monitoring & Logging (Task 8)

- **Logging:** every request/response is logged by the FastAPI app (structured stdout logs, container-friendly).
- **Metrics:** `/metrics` exposes Prometheus counters/histograms: `api_requests_total`, `heart_predictions_total`, `predict_latency_seconds`.
- **Prometheus** scrapes the API; **Grafana** loads a pre-provisioned dashboard (`Heart Disease API Monitoring`) with request rate, prediction counts by class, and p50/p95 latency panels. All wired via `docker-compose.yml`.

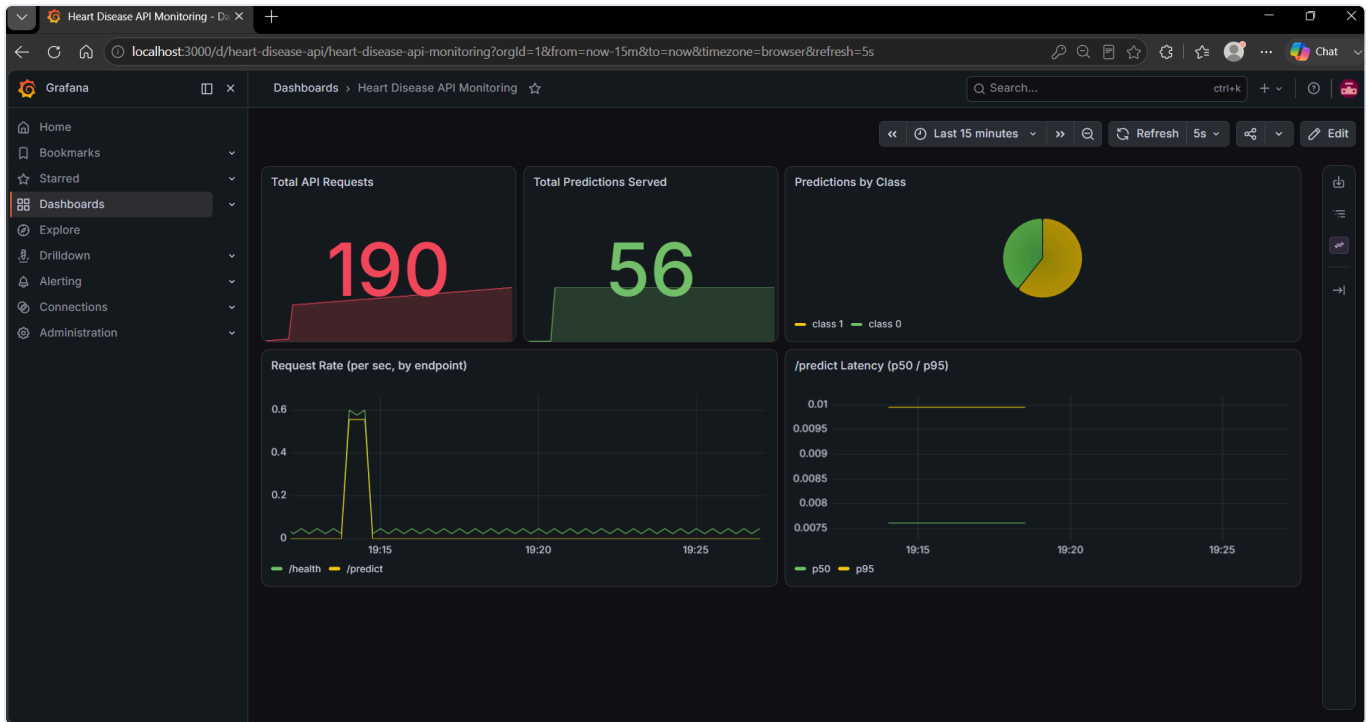


Figure 11: Grafana dashboard and Prometheus targets.

10. Setup / Install Instructions

See `README.md` for full commands. Summary:

```
pip install -r requirements.txt      # 1. environment
python data/download_data.py         # 2. data
python -m src.train                  # 3. train + MLflow
python -m pytest -v                 # 4. tests
uvicorn api.main:app --port 8000    # 5. serve (or docker compose up)
```

11. Repository & Deliverables

- **Repository:** <add your GitHub URL here>
- **Deployed API:** local Kubernetes / Docker (access via README sections 4 & 7)
- **Video walkthrough:** <add link here>

Deliverables checklist

- [x] Code, Dockerfile(s), requirements files
- [x] Cleaned dataset + download script
- [x] Notebooks (EDA, training, inference)
- [x] `tests/` with unit + API tests
- [x] GitHub Actions workflow YAML
- [x] Kubernetes manifests

- ☒ Monitoring (Prometheus + Grafana)
- ☒ Architecture diagram + report
- ☐ Screenshots (MLflow, CI, Docker, K8s, Grafana)
- ☐ Short pipeline video